MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	imgi_133_-288Wx360H-700135058-ecru-MODEL.jpg	**Target Domain:**	Face
Audit Timestamp:	2026-09-30 05:22:58 UTC	**SHA-256 Hash:**	dba6d19d45d974d31b895b97...
Forensic Verdict:	**VERIFIED AUTHENTIC MEDIA**	**Confidence:**	**58.0%**
Tamper Type:	Authentic Media Stream	**Bounding Box:**	None

Three-Stage Visual Forensics Pipeline



Stage 1: Source Image



Stage 2: Tamper Overlay



Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.5068 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 1.011
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.0279
MEFCN (Multi-Expert Consensus)	Face: 78.6% | Medical: 4.1% | Satellite: 6.1% | Regular: 5.6% | Natural: 5.6%
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 287x360 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 0.534 | Sensor Noise Var: 355.1

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.